Exercise 4

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R Lab: Data Structures - Vectors and Lists

Part A — Creating Vectors and Lists

Question #1:

A factory has a census of its workers. There are 50 workers in total. The following list shows their ages: `34, 28, 22, 36, 27, 18, 52, 39, 42, 29, 35, 31, 27, 22, 37, 34, 19, 20, 57, 49, 50, 37, 46, 25, 17, 37, 43, 53, 41, 51, 35, 24, 33, 41, 53, 40, 18, 44, 38, 41, 48, 27, 39, 19, 30, 61, 54, 58, 26, 18`.

a. Create a vector for the worker's age.

```
ages <- c(34, 28, 22, 36, 27, 18, 52, 39, 42, 29, 35, 31, 27, 22, 37, 34, 19, 20, 57, 49, 50, 37, 46, 25
ages
```

```
##  [1] 34 28 22 36 27 18 52 39 42 29 35 31 27 22 37 34 19 20 57 49 50 37 46 25 17
## [26] 37 43 53 41 51 35 24 33 41 53 40 18 44 38 41 48 27 39 19 30 61 54 58 26 18
```

b. How many data points?

```
length(ages)
```

```
## [1] 50
```

c. Find the minimum and the maximum value of worker's age.

```
min(ages)
```

```
## [1] 17
```

```
max(ages)
```

```
## [1] 61
```

d. Access the 13th to 18th elements and 33rd to 35th elements of worker's age.

```
ages[13:18]
```

```
## [1] 27 22 37 34 19 20
```

```
ages[33:35]
```

```
## [1] 33 41 53
```

```
ages[c(13:18, 33:35)]
```

```
## [1] 27 22 37 34 19 20 33 41 53
```

e. Access all but the 10th, 20th, 30th and 40th elements.

```
ages[-c(10, 20, 30, 40)]
```

```
##  [1] 34 28 22 36 27 18 52 39 42 35 31 27 22 37 34 19 20 57 50 37 46 25 17 37 43
## [26] 53 41 35 24 33 41 53 40 18 44 38 48 27 39 19 30 61 54 58 26 18
```

f. Modify the 4th, 14th, 24th, and 44th elements with value = 0.

```
ages_mod <- ages
ages_mod[c(4, 14, 24, 44)] <- 0
ages_mod
```

```
##  [1] 34 28 22  0 27 18 52 39 42 29 35 31 27  0 37 34 19 20 57 49 50 37 46  0 17
## [26] 37 43 53 41 51 35 24 33 41 53 40 18 44 38 41 48 27 39  0 30 61 54 58 26 18
```

Question #2:

R has actually lots of built-in datasets. For example, the `rivers` data "gives the lengths (in miles) of 141 'major' rivers in North America, as compiled by the US Geological Survey". Use `data(rivers)`.

a. Get the number of elements (`length`) in `rivers`, their sum (`sum`), mean (`mean`), median (`median`), variance (`var`), standard deviation (`sd`), minimum (`min`) and maximum (`max`).

```
data(rivers)

length(rivers)
```

```
## [1] 141
```

```
sum(rivers)
```

```
## [1] 83357
```

```
mean(rivers)
```

```
## [1] 591.1844
```

```
median(rivers)
```

```
## [1] 425
```

```
var(rivers)
```

```
## [1] 243908.4
```

```
sd(rivers)
```

```
## [1] 493.8708
```

```
min(rivers)
```

```
## [1] 135
```

```
max(rivers)
```

```
## [1] 3710
```

b. What is the result? The derived result must be in just one (1) vector name.

```
result <- c(length = length(rivers),
            sum    = sum(rivers),
            mean   = mean(rivers),
            median = median(rivers),
            var    = var(rivers),
```

```
            sd     = sd(rivers),
            min    = min(rivers),
            max    = max(rivers))
result
```

```
##      length         sum        mean      median         var          sd
##    141.0000  83357.0000    591.1844    425.0000 243908.4086    493.8708
##         min         max
##    135.0000   3710.0000
```

Question #3:

The table below gives the 25 most powerful celebrities and their annual pay as ranked by the editions of Forbes magazine and as listed on the Forbes.com website.

a. Create vectors according to the above table.

```
power_ranking <- 1:25

celebrity_name <- c("Tom Cruise", "Rolling Stones", "Oprah Winfrey", "U2", "Tiger Woods",
                    "Steven Spielberg", "Howard Stern", "50 Cent", "Cast of the Sopranos",
                    "Dan Brown", "Bruce Springsteen", "Donald Trump", "Muhammad Ali",
                    "Paul McCartney", "George Lucas", "Elton John", "David Letterman",
                    "Phil Mickelson", "J.K Rowling", "Bradd Pitt", "Peter Jackson",
                    "Dr. Phil McGraw", "Jay Lenon", "Celine Dion", "Kobe Bryant")

pay <- c(67, 90, 225, 110, 90, 332, 302, 41, 52, 88, 55, 44, 55,
         40, 233, 34, 40, 47, 75, 25, 39, 45, 32, 40, 31)
```

b. Modify the power ranking and pay of J.K. Rowling. Change power ranking to 15 and pay to 90.

```
power_ranking[19] <- 15
pay[19] <- 90

power_ranking
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 15 20 21 22 23 24 25
```

```
pay
```

```
##  [1]  67  90 225 110  90 332 302  41  52  88  55  44  55  40 233  34  40  47  90
## [20]  25  39  45  32  40  31
```

Question #4:

Create a list of ten (10) vegetables you ate during your lifetime. If none, just list down the vegetables in your mind.

```
veg <- list("Carrot", "Cabbage", "Tomato", "Eggplant", "Okra",
            "Squash", "Spinach", "Potato", "Onion", "Garlic")
veg
```

```
## [[1]]
## [1] "Carrot"
##
## [[2]]
## [1] "Cabbage"
```

```
##
## [[3]]
## [1] "Tomato"
##
## [[4]]
## [1] "Eggplant"
##
## [[5]]
## [1] "Okra"
##
## [[6]]
## [1] "Squash"
##
## [[7]]
## [1] "Spinach"
##
## [[8]]
## [1] "Potato"
##
## [[9]]
## [1] "Onion"
##
## [[10]]
## [1] "Garlic"
```

a. Add 2 additional vegetables after the last vegetable in the list. What is the result?

```
veg <- append(veg, list("Broccoli", "Cucumber"))
veg
```

```
## [[1]]
## [1] "Carrot"
##
## [[2]]
## [1] "Cabbage"
##
## [[3]]
## [1] "Tomato"
##
## [[4]]
## [1] "Eggplant"
##
## [[5]]
## [1] "Okra"
##
## [[6]]
## [1] "Squash"
##
## [[7]]
## [1] "Spinach"
##
## [[8]]
## [1] "Potato"
##
## [[9]]
```

```
## [1] "Onion"
##
## [[10]]
## [1] "Garlic"
##
## [[11]]
## [1] "Broccoli"
##
## [[12]]
## [1] "Cucumber"
```

b. How many data points do you have?

```
length(veg)
```

```
## [1] 12
```

c. Add 4 additional vegetables after index 5. What is the result?

```
veg <- append(veg, list("Radish", "Pumpkin", "Lettuce", "Celery"), after = 5)
veg
```

```
## [[1]]
## [1] "Carrot"
##
## [[2]]
## [1] "Cabbage"
##
## [[3]]
## [1] "Tomato"
##
## [[4]]
## [1] "Eggplant"
##
## [[5]]
## [1] "Okra"
##
## [[6]]
## [1] "Radish"
##
## [[7]]
## [1] "Pumpkin"
##
## [[8]]
## [1] "Lettuce"
##
## [[9]]
## [1] "Celery"
##
## [[10]]
## [1] "Squash"
##
## [[11]]
## [1] "Spinach"
##
## [[12]]
```

```
## [1] "Potato"
##
## [[13]]
## [1] "Onion"
##
## [[14]]
## [1] "Garlic"
##
## [[15]]
## [1] "Broccoli"
##
## [[16]]
## [1] "Cucumber"
```

d. How many data points does your vegetable list have?

```
length(veg)
```

```
## [1] 16
```